

Vijayeshnanda Vasanthakumar

Aspiring computational physicist with strong foundations in classical mechanics, data-driven modeling, and software development. Demonstrated leadership in STEM education and original research experience in physics-based simulation and machine learning.

RESEARCH AND WORK EXPERIENCE

Independent Research

Dec 2025 - Present

Conducting independent computational astrophysics research using data-driven methods to identify rare stellar outliers in Gaia DR3. Going to use algorithmic modeling, statistical analysis, and simulation to large astronomical datasets, translating physical assumptions into quantitative code to study stellar evolution and extreme astrophysical phenomena

VDart — Web Development Intern

Jun 2024 - Sept 2024

Interned at VDart, a rising tech consulting company. Assisted with software development of VDart website (search engine), gaining valuable experience in Wordpress and phpMyAdmin, and exposed to professional environment.

Hackclub — Developer/Hackathon Participant

Feb 2025 - Present

Designed and built multiple software projects focused on machine learning, AI systems, and human-computer interaction. Collaborated in fast-paced hackathon environments and helped organize local hackathons. Honorable Mention out of 113 projects in an online international high school hackathon hosted from Manteca

OTHER EXPERIENCE

EHS (Astro)Physics Club — President (Astrophysics)

Jan 2024 - Present

Conducted educational sessions to prepare for both the Physics Olympiad (kinematics, FBDs, Newton's laws, and trigonometry) and the Astronomy/Astrophysics Olympiad (lightshift, stars, and space time continuum). IAAC Final Round Qualifier.

EHS Science Olympiad — Secretary

Aug 2023 - Present

Participated in 5+ national satellite tournaments and 5+ regional tournaments, mostly consistent top 25% performance in my events of cryptography/robotics.

EETHAL Foundation — Young Translator, English Teacher

Aug 2023 - Present

Translated kids stories from English to Tamil using the Storyweaver website;. Tutored classes of 20-30 kids in India, focusing on their English.

Young American Patriots — Lead Drummer

Aug 2024 - Present

Performed in multiple different parades and ceremonies to represent the old Revolutionary War music.

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SKILLS

Physics: Mechanics, Kinematics, Astrophysics, Quantitative modeling

Comp Sci: Python, Machine Learning, Data analysis, Web Development, Databases

ACADEMIC EXPERIENCE

Emerald High: Developing strong academic foundations and growth through extracurriculars and school

Art of Problem Solving:

Developed critical thinking and problem solving skills through rigorous work

HONORS AND AWARDS

International Astronomy and Astrophysics Competition
Final Round Qualifier

Presidential Volunteer Service Award

Honorable Mention at Teca Hacks

Certificate of Excellence from the California Chapter of the Daughters of the American Revolution

4th place in Math/Astronomy/Physics category at Alameda County Science and Engineering Fair

Certificate of Special Congressional Recognition for the Congressional App Challenge